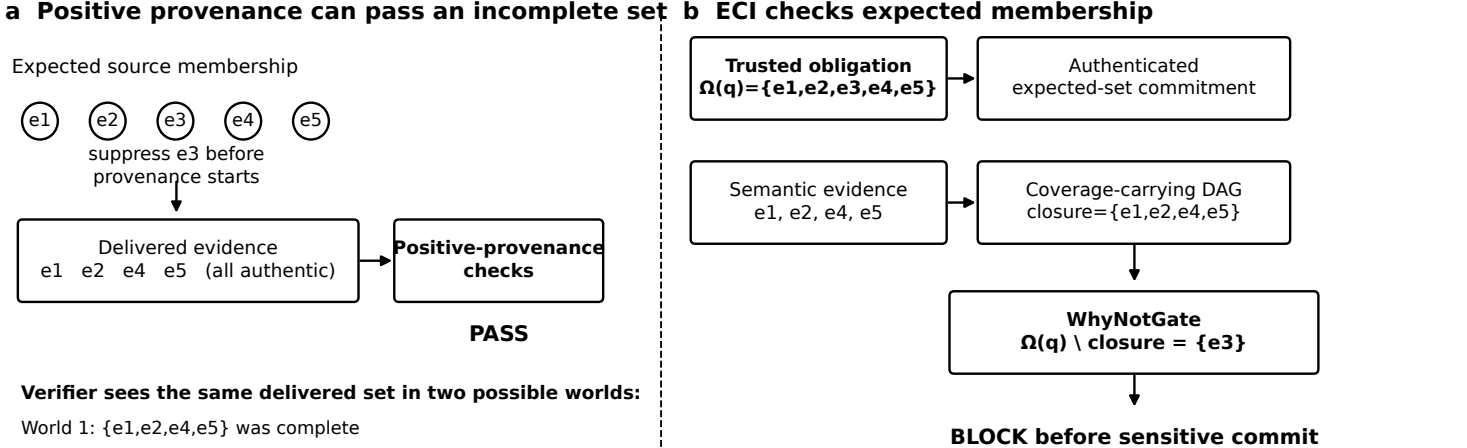

a Positive provenance can pass an incomplete set
Expected source membership
e1
e2
e3
e4
e5
suppress e3 before provenance starts
Delivered evidence
e1 e2 e4 e5 (all authentic)
Positive-provenance checks
PASS
Verifier sees the same delivered set in two possible worlds:
World 1: {e1,e2,e4,e5} was complete
World 2: e3 existed but disappeared upstream
Observed evidence alone cannot tell which world is real.
b ECI checks expected membership
Trusted obligation
Ω(q)={e1,e2,e3,e4,e5}
Authenticated expected-set commitment
Semantic evidence
e1, e2, e4, e5
Coverage-carrying DAG
closure={e1,e2,e4,e5}
WhyNotGate
Ω(q) \ closure = {e3}
BLOCK before sensitive commit
Negative provenance asks what should have been present but is missing.